

Practical Exercise 2: Circuit Design

PART A

a) Series Circuit

1.) Calculate Total resistance.

$$R_T = R_1 + R_2 + R_3$$

$$R_T = 20 + 30 + 50 = \underline{\underline{100\Omega}}$$

2.) Calculate Total current

$$I = \frac{125}{100} = \underline{\underline{1.25\text{ A}}}$$

3) Current through each resistor

$$\text{Current through } 20\Omega = 1.25\text{ A}$$

$$\text{'' '' } 30\Omega = 1.25\text{ A}$$

$$\text{'' '' } 50\Omega = 1.25\text{ A}$$

$$\Rightarrow \text{All resistor currents} = \underline{\underline{1.25\text{ A}}}$$

4) Voltage Drop Across Each Resistor

$$V = IR$$

20 Ω resistor

$$V = 1.25 \times 20 = \underline{\underline{25\text{ V}}}$$

30 Ω resistor

$$V = 1.25 \times 30 = \underline{\underline{37.5\text{ V}}}$$

50 Ω resistor

$$V = 1.25 \times 50 = \underline{\underline{62.5\text{ V}}}$$

5) Calculate Power for each resistor

$$P = VI$$

20 Ω resistor

$$P = 25 \times 1.25 = \underline{\underline{31.25\text{ W}}}$$

30 Ω resistor

$$37.5 \times 1.25 = \underline{\underline{46.875\text{ W}}}$$

50 Ω resistor

$$62.5 \times 1.25 = \underline{\underline{78.125\text{ W}}}$$

b) Parallel Circuit

1) Calculate Total Resistance

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$= 0.05 + 0.01 + 0.02$$

$$= 0.08$$

$$R_T = \frac{1}{0.08} = \underline{\underline{12.5 \Omega}}$$

2) Calculate Total current

$$I = \frac{V}{R} = \frac{125}{12.5}$$

$$I = \underline{\underline{10 A}}$$

3) Current through each resistor

$$I = \frac{V}{R}$$

20 Ω resistor :

$$I = \frac{125}{20} = 6.25 A$$

100 Ω resistor :

$$I = \frac{125}{100} = 1.25 A$$

50 Ω resistor :

$$I = \frac{125}{50} = 2.5 A$$

4) Voltage Drop Across Each Resistor

Voltage across each resistor
= source voltage

$$20 \Omega \Rightarrow 125 V$$

$$100 \Omega \Rightarrow 125 V$$

$$50 \Omega \Rightarrow \underline{\underline{125 V}}$$

5) Power Dissipation.

$$P = VI$$

20 Ω resistor :

$$P = 125 \times 6.25 = \underline{\underline{781.25 W}}$$

100 Ω resistor :

$$P = 125 \times 1.25 = \underline{\underline{156.25 W}}$$

50 Ω resistor :

$$P = 125 \times 2.5 = \underline{\underline{312.5 W}}$$